

5. Hydrogen peroxide, H_2O_2 , has anti-fungal properties and can be used to preserve milk. Before the milk is used to make cheese the H_2O_2 needs to be removed.

Hydrogen peroxide decomposes to form water and oxygen gas as shown below.

hydrogen peroxide $\rightarrow$ water + oxygen

- (a) Write a balanced symbol equation for the reaction. [2]

.....

- (b) The rate of this reaction is very slow at room temperature without a catalyst.

A student suggests that the rate of reaction can be increased by adding a catalyst or increasing the temperature.

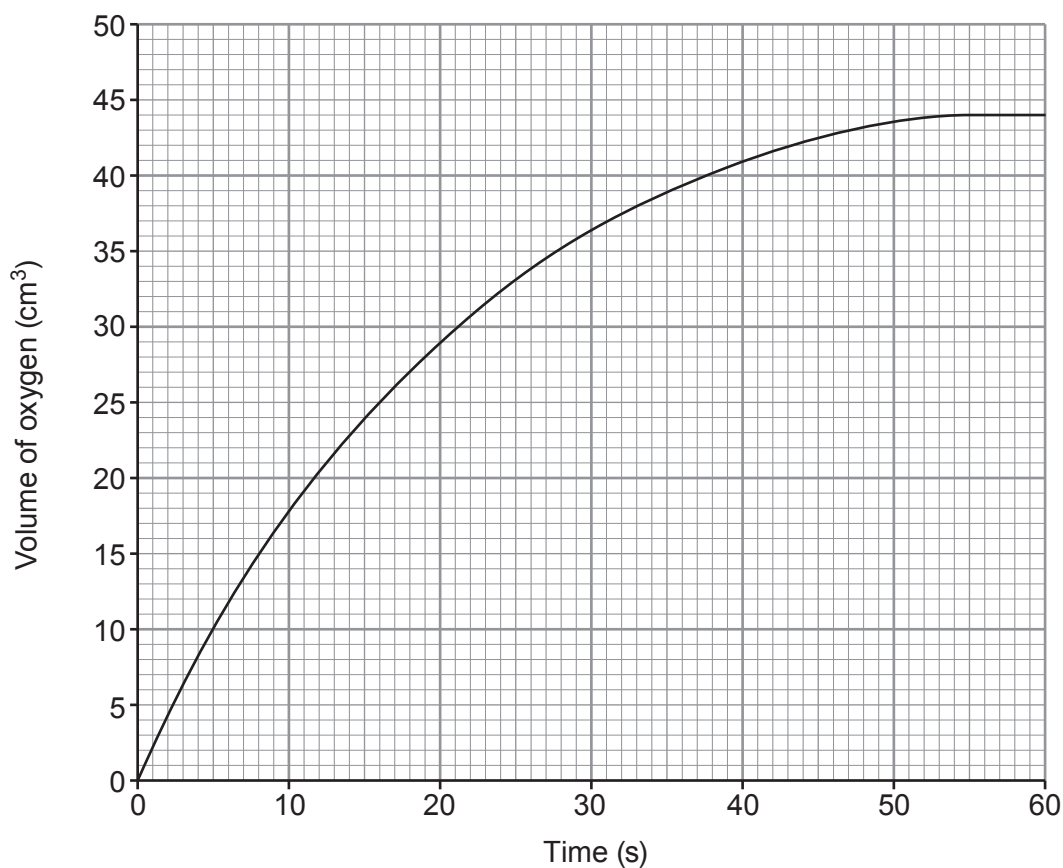
State **two** variables that should be controlled in an experiment to determine the effect of temperature on the rate of decomposition of hydrogen peroxide. [2]

.....

.....

- (c) In an investigation, the student measured the volume of oxygen collected in a measuring cylinder every 10 seconds for a total of 60 seconds.

The graph below shows the results at 22°C.



- (i) Calculate the rate of reaction at 30 seconds by drawing a tangent. [3]

Rate = cm³/s

- (ii) The rate of reaction at 5 seconds was 2 cm³/s. Explain, using particle theory, why the rate of decomposition was less at 30 seconds. [2]

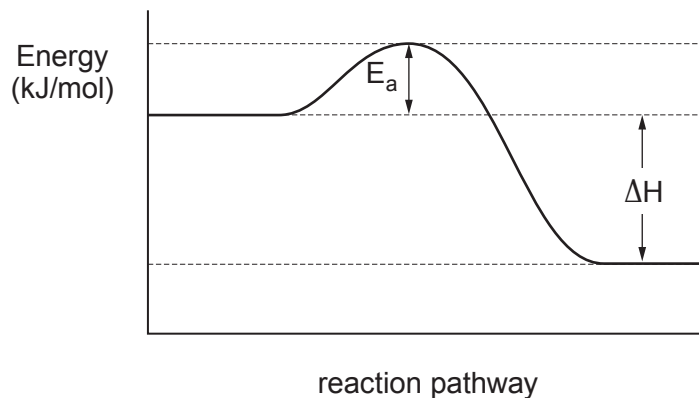
.....

.....

.....

- (iii) The experiment was repeated at 35°C. **Add** a line to the grid to show the results you would expect. [2]

- (d) The energy changes during the decomposition of hydrogen peroxide without a catalyst are shown below. The energy E_a is 75 kJ/mol and ΔH is -196.1 kJ/mol.



- (i) Explain what type of reaction is shown by this energy diagram. [2]

.....

.....

- (ii) A student states that the values of E_a and ΔH do not change when a catalyst is used. Explain whether you agree with the student. [3]

.....

.....

.....

.....

- (e) The rate of a chemical reaction doubles if the temperature is increased by 10°C. At 6°C, milk undergoes a chemical reaction that makes it go sour in 6 days. Calculate how long it will take milk to go sour at 36°C. [2]

Time =